

Experiment - 2

Aim - Convert the E-R diagram into a relational schema write complete MYSQL CREATE TABLE statements with primary keys, foreign keys, NOT Null, UNIQUE, ON DELETE CASCADE & SET NULL constraints.
- Insert sample data demonstrate referential integrity violations.

Source Code -

```
DROP DATABASE IF EXISTS ecommerce_db;  
CREATE DATABASE ecommerce_db;  
USE ecommerce_db;
```

— 1) CUSTOMER (Strong entity)

```
CREATE TABLE Customer(  
    customer_id INT AUTO_INCREMENT PRIMARY KEY  
    first_name VARCHAR(50) NOT Null,  
    last_name VARCHAR(50) NOT Null,  
    email VARCHAR(100) NOT Null,  
    ) ENGINE = InnoDB;
```

— Multi-valued attribute: Customer phone Numbers.

```
CREATE TABLE Customer_Phone(  
    customer_id INT NOT Null,  
    phone VARCHAR(15) NOT Null,  
    PRIMARY KEY (customer_id, phone),  
    FOREIGN KEY (customer_id) REFERENCES customer(customer_id)  
    ON DELETE CASCADE  
    ) ENGINE = InnoDB;
```

2.) ADDRESS (weak entity - depends on Customer)

CREATE TABLE Address (

customer_id INT NOT Null,

address_id INT NOT Null, --- partial Key

street VARCHAR(100) NOT Null,

city VARCHAR(50) NOT Null,

state VARCHAR(50) NOT Null,

pincode VARCHAR(10) NOT Null,

PRIMARY KEY (customer_id, address_id),

FOREIGN KEY (customer_id) REFERENCES

Customer(customer_id) ON DELETE CASCADE

--- Composite
Key of weak
entity.

3.) CATEGORY (Strong Entity)

CREATE TABLE Category (

Category_id INT AUTO-INCREMENT PRIMARY KEY,

Category_name VARCHAR(50) NOT Null UNIQUE.

) ENGINE = INNODB

4.) SELLER (Strong Entity)

CREATE TABLE Seller (

Seller_id INT AUTO-INCREMENT PRIMARY KEY,

Seller_name VARCHAR(100) NOT Null,

Email VARCHAR(100) NOT Null UNIQUE

) ENGINE = INNODB;

CREATE TABLE Seller_Phone (

Seller_id INT NOT Null,

phone VARCHAR(15) NOT Null,

PRIMARY KEY (Seller_id, phone)

FOREIGN KEY (Seller_id) REFERENCES Seller(Seller_id)

ON DELETE CASCADE

) ENGINE = INNODB;

2) PRODUCT (Strong entity, with specialization)

CREATE TABLE PRODUCT(

product_id INT AUTO_INCREMENT PRIMARY KEY,

product_name VARCHAR(150) NOT NULL,

price DECIMAL(10,2) NOT NULL,

stock INT NOT NULL DEFAULT 0,

category_id INT NULL,

seller_id INT NOT NULL,

FOREIGN KEY(category_id) REFERENCES category(
category_id)

ON DELETE SET NULL,

FOREIGN KEY(seller_id) REFERENCES seller(seller_id)

ON DELETE CASCADE

) ENGINE = InnoDB;

- Specialization: ~~Electronics~~ (disjoint subtype of Product)

CREATE TABLE Electronics(

product_id INT PRIMARY KEY,

warranty_period INT,

brand VARCHAR(50),

FOREIGN KEY(product_id) REFERENCES Product(
product_id)

ON DELETE CASCADE

) ENGINE = InnoDB;

- Specialization: clothing (disjoint subtype of Product)

```
CREATE TABLE clothing (  
    product_id INT PRIMARY KEY,  
    size VARCHAR(10),  
    fabric VARCHAR(50),  
    FOREIGN KEY (product_id) REFERENCES Product  
        (product_id)  
        ON DELETE CASCADE  
);
```

ENGINE = InnoDB;

6.) ORDER (Strong Entity) - "order" is a reserved word in MySQL.

```
CREATE TABLE orders (  
    order_id INT AUTO_INCREMENT PRIMARY KEY,  
    order_date DATE NOT NULL,  
    total_amount DECIMAL(10,2) NOT NULL,  
    customer_id INT NOT NULL,  
    FOREIGN KEY (customer_id) REFERENCES  
        Customer(customer_id)  
        ON DELETE CASCADE  
);
```

ENGINE = InnoDB;

7.) ORDER_ITEM (Weak entity - depends on Orders)

```
CREATE TABLE OrderItem (  
    order_id INT NOT NULL,  
    item_no INT NOT NULL, --- partial key  
    quantity INT NOT NULL,  
    unit_price DECIMAL(10,2) NOT NULL,  
    product_id INT NULL,  
    PRIMARY KEY (order_id, item_no), --- composite  
key of weak entity.  
    FOREIGN KEY (order_id) REFERENCES orders (  
        order_id)  
        ON DELETE CASCADE,  
);
```

```
    FOREIGN KEY (product_id) REFERENCES Product  
        (product_id) ON DELETE SET NULL  
);
```

ENGINE = InnoDB;

8) PAYMENT (Strong entity, 1:1 with orders)

```
CREATE TABLE Payment (  
    payment_id INT AUTO-INCREMENT PRIMARY KEY,  
    amount DECIMAL (10,2) NOT NULL,  
    payment_date DATE NOT NULL,  
    payment_mode VARCHAR (20) NOT NULL,  
    order_id INT NOT NULL UNIQUE, -- (UNIQUE enforces  
    FOREIGN KEY (order_id) REFERENCES orders (order_id) the 1:1 cardinality)  
    ON DELETE CASCADE  
); ENGINE = InnoDB;
```

9) DELIVERY (Strong entity, 1:1 with orders)

```
CREATE TABLE Delivery (  
    delivery_id INT AUTO-INCREMENT PRIMARY KEY  
    delivery_date DATE,  
    status VARCHAR (20) NOT NULL,  
    tracking_number VARCHAR (30) UNIQUE,  
    order_id INT NOT NULL UNIQUE,  
    FOREIGN KEY (order_id) REFERENCE (order_id)  
    ON DELETE CASCADE
```

)ENGINE = InnoDB;

-- SAMPLE DATA →

```
INSERT INTO CUSTOMER (first_name, last_name, email)  
VALUES  
( 'Akhil', 'Verma', 'Akhil.Verma@example.com'),  
( 'Priya', 'Iyer', 'priya.cyber@example.com');
```

```
INSERT INTO Customer_Phone (customer_id, phone) VALUES  
( 1, '9696 182681'),  
( 2, '9455386431'),  
( 2, '8691553864');
```

INSERT INTO Address (customer_id, address_id, Street, city, State, pincode) VALUES

(1, 1, '42 MG Road', 'Barabanki', 'Uttar Pradesh', '226001')
(2, 1, '45 Anna Salai', 'chennai', 'Tamil Nadu', '600002');

INSERT INTO CATEGORY (category_name) VALUES
('Electronics'), ('clothing'), ('Groceries');

INSERT INTO Seller (seller_name, email) VALUES
('Tech Bazar Pvt Ltd', 'contact@techbazar.com'),
('FashionHub India', 'support@fashionhub.com');

INSERT INTO Seller_Phone (seller_id, phone) VALUES
(1, '011-40001234'),
(2, '022-50005678');

INSERT INTO Product (product_name, price, stock, category_id, seller_id) VALUES
('Wireless Earbuds X200', 2499.00, 50, 1, 1),
('cotton Kurta-Blue', 899.00, 100, 2, 2);

INSERT INTO CLOTHING (product_id, size, fabric) VALUES
(2, 'M', 'cotton');

INSERT INTO orders (order_date, total_amount, customer_id) VALUES
('2026-08-02', 3388.00, 1);

INSERT INTO Electronics (product_id, warranty_period, brand) VALUES
(1, 12, 'soundwave');

INSERT INTO order_item (order_id, item_no., quantity, unit_price, product_id) VALUES
(1, 1, 1, 2499.00, 1),
(1, 2, 1, 899.00, 2);

INSERT INTO PAYMENT (amount, payment_date, payment_mode,
order_id) VALUES

(3398.00, '2026-08-01', 'UPI', 1);

INSERT INTO Delivery (delivery_date, status,
tracking_number, order_id) VALUES
(2026-08-05, 'Shipped', 'TRK123456789', 1);

DEMONSTRATING REFERENTIAL INTEGRITY

a) INSERT violation: order_id 999 don't exist in orders.

Expected: Error 1452 - Can't add or update a child row:
a foreign key constraint fails (order_id FK)

INSERT INTO order-Item (order_id, item_no, quantity,
unit_price, product_id)

VALUES (999, 1, 2, 500.00, 1);

b) INSERT violation: product_id 999 don't exist in product.

Expected: Error 1452 - foreign key constraint fails
(product_id FK)

INSERT INTO order-Item (order_id, item_no, quantity,
unit_price, product_id)

values (1, 3, 1, 100.00, 999);

c) UNIQUE violation: This order already has a
payment (1:1 constraint)

Expected: Error 1062 - Duplicate entry '1' for key
'order_id'.

INSERT INTO Payment (amount, payment_date,
payment_mode, order_id)

values (500.00, '2026-08-02', 'Cash', 1);

ON DELETE SET NULL demonstration: delete a
category.
Expected: product category_id for its products
becomes NULL, rows survive

2) - Delete FROM Category WHERE category_id=1;
- SELECT * FROM Product WHERE product_id=1;
category_id is now NULL.

e) ON DELETE CASCADE demonstration: delete a
- expected: their Address, Customer-Phone, orders
rows are deleted
- automatically, when in turn cascades to
order-item,
- Payment, & delivery for those address

- DELETE FROM Customer WHERE customer_id=1;
- SELECT * FROM orders WHERE customer_id=1; empty result
- SELECT * FROM Payment WHERE order_id=1; empty result

f) NOT NULL violation: order_date is required
- Expected: Error 1364 - field 'order_date' doesn't
have a default value.

- Insert INTO orders (total_amount, customer_id)
VALUES
(50000, 1);

~~A 26/08/26~~